



Structured Codebook Based Hierarchical Framework for DNN for Computationally Efficient Speech Enhancement

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Abstract

Deep Neural Networks (DNN) based single-channel speech enhancement techniques have surpassed traditional techniques in handling non-stationary noise, however, they are computationally demanding. In this work, we introduce a novel Hierarchical Framework for DNNs (HF-DNN) for speech enhancement that replaces a single complex and computationally expensive DNN model with multiple simpler and less complex DNNs that are hierarchically connected. This is achieved by using structured codebooks of speech parameters, like log power spectra, that are generated by exploiting hierarchical relation between the speech training data. The proposed HF-DNN reduces computational complexity significantly compared to a large DNN while maintaining speech enhancement performance. Importantly, such a framework can be extended to other speech processing tasks, such as speech recognition, speaker verification, etc., where parametric models of speech data are utilized.

Index Terms: Hierarchical DNN, speech enhancement, codebook

1. Introduction

Speech enhancement refers to the removal of noise from the noisy speech signals to provide a cleaner speech signal. It has applications in several fields such as mobile communication, hearing aids, preprocessing for speech recognition, etc. Many single-channel speech enhancement algorithms have been developed in the past few decades [1, 2]. Among these, the model-based approaches employing trained models such as hidden Markov models (HMM), Gaussian mixture models (GMM), codebooks, DNNs, etc., have shown better performance in non-stationary noise conditions by exploiting prior information of speech and noise data. DNN model-based approaches have gained significance recently due to their ability to capture the nonlinear mapping between noisy and corresponding clean signals efficiently. Several DNN architectures have been used for the purpose of single-channel speech enhancement [2].

DNN-based approaches are advantageous in terms of better speech enhancement, however, their performance comes at the cost of higher computational complexity [3], generally measured in terms of the number of trainable parameters of the model. In this work, a novel Hierarchical Framework for DNNs (HF-DNN) based on a structured codebook of clean speech data is introduced to reduce the computational demand of DNNs. The framework allows for the reduction of the computational cost of DNNs while maintaining and at times even improving, the speech enhancement performance compared to a single large DNN.

The idea of Hierarchical DNNs (HDNNs) was introduced

in [4] wherein a multilayer network structure was used. In this structure, the top-level layers extracted coarse features whereas the subsequent layers acted as finer classifiers. This approach has been applied in the field of image processing to address complex classification challenges. In [5], Koo et.al., developed a novel architecture for hierarchical image classification by integrating Convolutional Neural Networks (CNNs) as feature learners with Recurrent Neural Networks (RNNs) and Sequence-to-Sequence (S2S) models. The applicability of HDNN for image processing is mainly due to the possibility of breaking the image analysis task into subtasks of different complexity at multiple levels but related hierarchically. A similar approach has not been explored in the case of input speech data.

Noorani et. al. [6] proposed a Hierarchical DNN for speech separation wherein the hierarchy represents different stages of processing input data in terms of gender classification, source separation and enhancement but not by exploiting any hierarchical relation in the prior speech data. The hierarchical framework introduced in this paper for speech enhancement is different from the above mentioned approaches as it is built using structured codebooks generated by utilising the hierarchical relation of the speech training data.

The remainder of the paper is organized as follows. The related background work of the structured codebook based speech enhancement is described in Section 2. In Section 3, the proposed structured codebook based Hierarchical Framework for DNNs is described. The details and results of the experiments performed for speech enhancement using the proposed HF-DNN are presented in Section 4. Finally, Section 5 provides the conclusions.

2. Related Background Work

The structured codebook of speech spectra was introduced in [7] to reduce the computational complexity of the codebook based speech enhancement (CBSE) methods [8]. The CBSE methods employed trained Linear Predictive Coding (LPC) codebooks of speech and noise data to estimate the power spectral density (PSD) of the input noisy frames for the purpose of speech enhancement. The signal model assumed was that of additive noise. The CBSE scheme required calculations which involved every possible pair of LPC vectors from the speech and noise LPC codebooks. This led to a significant amount of computation as the number of speech codebook vectors needed to be large to represent a variety of speech spectral shapes accurately. However, in [7], it was observed that only a few of these pairs contributed significantly to the estimation of a given observed noisy frame. To restrict computations to only relevant speech codebook vectors for a given noisy segment, the struc-

tured codebook was proposed, which imposed a tree structure on the speech codebook, as described below.

Consider a speech LPC codebook of size 256, trained with the Linde-Buzo-Gray (LBG) [9] algorithm using the Itakura-Saito (IS) measure [10] as the distortion measure. Each entry in the codebook generated is a centroid of a corresponding partition of the speech training dataset, i.e., all the training vectors which are closest to that centroid than any other centroid are part of the partition or cluster represented by that centroid. It could also be observed that two LPC vectors which are part of one cluster are likely to be closer to each other (in the IS sense) than the vectors from other clusters. Thus, vectors from one cluster are likely to either match or mismatch with the clean signal part of the input noisy segment.

To build a structured codebook, the 256 centroid codebook vectors are used as a training dataset to generate 32 new centroid codebooks using the LBG algorithm again. The 256 entries formed the bottom-most level (Level 0) in the structured codebook, as shown in Figure 1. The Level 1 nodes divided the 256 vectors of Level 0 into 32 clusters and constituted the next higher level. Extending the same procedure to the next level, 4 nodes are generated out of the 32 centroids of Level 1 and added to the structured codebook to constitute the Level 2.

During the enhancement, the tree is traversed in a top-down order. First, for a given input noisy frame, the Level 2 speech LPC vectors are considered for pairing with the noise codebook vectors to estimate the noisy PSD. For each such pair, the likelihood of that pair having generated the observed noisy frame is computed using the log spectral distortion (LSD) measure as in the CBSE method [8]. Next, the likelihood computation is repeated for Level 1 speech LPC vectors but only for those child nodes at Level 1 whose parent node at Level 2 resulted in the highest likelihood. The same procedure for likelihood computation is extended to the next lower level. The child nodes of the highest likelihood node at the penultimate level, i.e., Level 1 in Figure 1, formed the Most Likelihood Cluster (MLC) for the observed noisy segment. As the vectors in MLC are close to their centroids, which is the highest likelihood node of the penultimate level, hence, they are expected to be closer to the clean signal part of the input noisy frame than the vectors from the other clusters. This way, all pairs are not involved in computation and only MLC vectors are considered leading to computational cost reduction.

While the structured codebook results in significant computational reduction, the enhancement of noisy speech is only marginally degraded compared to the original CBSE scheme. Further, structured codebooks of different numbers of levels and nodes in each level can be considered for better performance. Also, instead of selecting the child nodes of just the topmost likelihood parent node, child nodes from the top two or more likelihood parent nodes could also be considered at every level of the structured codebook as a trade-off between computational cost and speech enhancement performance.

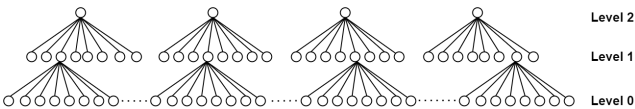


Figure 1: Three-Tier structured codebook constructed in a bottom-up fashion.

3. Proposed Structured Codebook Based Hierarchical DNN Framework

In this section, we propose a hierarchical framework for DNN, HF-DNN, using a structured codebook for speech enhancement. The proposed framework consists of two types of DNNs: DNN classifiers and speech enhancement DNNs (SE-DNNs). We propose HF-DNN for input as a single frame as well for segments consisting of group of frames.

3.1. HF-DNN for Single frame input

Consider a two-tier structured codebook created from clean speech LPS vectors. The topmost layer of this structured codebook contains four classes, while the bottom layer encompasses a total of 32 subclasses. For a given noisy LPS vector and its corresponding clean LPS, the clean LPS is compared with the nodes of the structured codebook using the minimum LSD, which identifies the closest centroids at each codebook layer. This comparison traces a defined path from the topmost to the bottom layer of the structured codebook. The HF-DNN is constructed to mirror this trajectory for the corresponding noisy LPS vector.

Figure 2 depicts a schematic of the HF-DNN, where training begins at the topmost level, Level 2. At this level, a DNN classifier is trained to categorize all incoming noisy LPS vectors into four classes. Each of these four classes is then further subdivided by a second DNN classifier at Level 1, which categorizes each noisy LPS vector into one of 32 subclasses. It is important to note that this training utilizes the structured codebook trajectory as previously explained.

As the training process moves from higher to lower levels, the amount of noisy data required for each subsequent DNN classifier reduces. Consequently, the DNN classifiers within the Hierarchical Framework at Levels 2 and 1 are simpler and less complex than a single, comprehensive DNN designed for speech data. At the lowest level, Level 0, the classified noisy LPS vectors are used to train the SE-DNNs to produce clean speech LPS vectors, as illustrated in Figure 2. These SE-DNNs are significantly simpler and less computationally demanding because they manage considerably less training data.

During testing, the HF-DNN processes input noisy speech vectors starting from the top layer. Initially, the Level 2 DNN classifier assigns an appropriate Level 1 class to the input noisy LPS vector. This noisy LPS vector is then further classified by the corresponding Level 1 DNN into one of the classes at Level 0. Subsequently, at Level 0, only the relevant SE-DNN is activated to provide the clean speech LPS estimate and not other SE-DNNs.

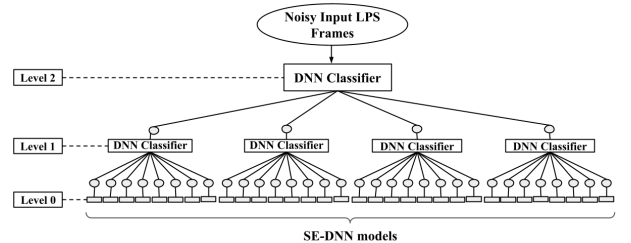


Figure 2: Three-Tier Hierarchical DNN framework for speech enhancement.

3.2. HF-DNN for Segment of frames as input

The codebooks generated for a parametric form of speech frame are limited by duration of the frame to which the parametrisation is possible. For example, LPC parametrisation is permissible only for frames below duration 32 ms. In order to cater to DNNs which process segments spanning multiple frame duration, the HF-DNN can be built by classifying input segments based on distribution of number of its frames from a particular codebook cluster. Consider, for example, a codebook with four entries at Level 1 in Figure 2. The input segment can then be classified based on the top codebook cluster or top two codebook clusters out of the 4 codebook clusters, to which most of the frames of that segment belong to.

4. Experimental Results

In this section, we present the results of the experiments that were performed to test the performance of the proposed HF-DNN framework for the case of input as segment of frames.

4.1. Training of Structured Codebook

An LPS structured codebook consisting of 4 codebooks entries was created using k-means clustering applied to clean speech data from the VoiceBank corpus [11]. The experimental training data consisted of 11,572 clean utterances, each lasting between 5 to 7 seconds, from a total of 28 speakers. LPS vectors were extracted from the speech files using a Hanning window of 400 samples, with a 50% overlap and a sampling frequency of 16 kHz. The structured codebook consisted of one top level of 4 nodes.

4.2. Reference DNN

In this study, the state-of-the-art SEMamba was employed as the Reference DNN as provided in [12]. The SEMamba architecture relies on the Mamba model, a state-space model designed for efficient sequence handling. This architecture comprises a series of convolutional layers combined with state-space blocks, which dynamically adjust to the input data through a selective state-space mechanism. SEMamba uses four convolutional layers to extract features from the input spectrogram. After initial feature extraction, the architecture employs a state-space model to process temporal dependencies and sequence dynamics effectively. This adaptability allows SEMamba to enhance speech signals by focusing precisely on relevant audio details, optimizing noise suppression and signal clarity.

The training dataset for the Reference DNN architecture consisted of 11,572 noisy utterances at SNRs of 0 dB, 5 dB, 10 dB, 15 dB and 20 dB. These utterances were obtained from the VoiceBank-DEMAND dataset [13] and spanned 28 speakers. The SEMamba architecture was trained over 200 epochs.

4.3. Training of DNN classifiers for HF-DNN

Multi-Layer Perceptron (MLP) with three fully connected layers was employed for the DNN classifiers. It comprised two hidden layers with ReLU activations. The model uses the Adam optimizer for training, and it concludes with two dense output layers, each applying a softmax activation function for multi-class classification. The classifier was trained using cross-entropy loss. The DNN classifier is the key component of the HF-DNN framework, with its primary role being to classify noisy speech frames into predefined cluster labels derived from a structured codebook. It comprised total number of 50K train-

able parameters. The training dataset for the DNN classifiers also consisted of the same 11,572 noisy utterances, as explained in the above subsection, and frames were extracted from these noisy files using a Hanning window of 400 samples, with 50% overlap and a sampling frequency of 16 kHz. The DNN classifier was trained over 30 epochs.

Once all frames in a 2-second segment are classified, the distribution of predicted labels is analyzed to determine the dominant (primary) and second-most frequent (secondary) cluster labels, as discussed in the Subsection 3.2. For every 2-second noisy segment, the DNN classifier assigns primary and secondary classes, resulting in 12 such classifications.

The DNN classifier was evaluated against test noisy files using 180 clean files from a male speaker from the Wall Street Journal (WSJ) speech data [14], which were not included in any of the other DNNs trained for these experiments. The testing phase involved adding five different challenging noise types from the NOISEX database [15] to clean speech across seven SNR levels, ranging from -10 dB to 20 dB with an interval of 5 dB. When the DNN classifier was tested against these test files, the accuracy, when compared to true segment classification, was 88%.

4.4. Training of speech enhancement DNN for HF-DNN

For SE-DNN, SEMambas were utilized. They were also trained on the same noisy utterances and SNR levels as mentioned in Subsection 4.2. In order to train these SEMambas, the training data segments were classified based on the top 2 codebook clusters. For a 4-entry codebook, this would require 12 SE-DNNs to be constructed, representing 12 classes of top 2 clusters. Each of these SEMambas had around 21K trainable parameters, and they were trained across 200 epochs as the baseline SEMamba.

4.5. Speech Enhancement Experimental Results

In order to test the proposed HF-DNN framework, only SE-DNNs need to be tested. The same test noisy files that were used in the DNN classifier were used. The test noisy files were enhanced using the proposed HF-DNN, the Reference DNN, and other state-of-the-art DNN speech enhancement algorithms, namely MANNER [16] and MP-SENet [17]. Apart from this, a model compression technique like Knowledge Distillation [18] was also applied to the Reference DNN, similar to the HF-DNN framework.

The output was evaluated using six performance metrics: Perceptual Evaluation of Speech Quality (PESQ) [19], Short-Term Objective Intelligibility (STOI) [20], Segmental SNR (SSNR), Continuous Speech Intelligibility Grade (CSIG), Continuous Background Audio Knowledge (CBAK) and Continuous Overall Voice Likability (COVL) [21]. Additionally, computational efficiency was assessed using the number of trainable parameters (Para.) and multiply-accumulate operations (MACs), which serve as indicators of the feasibility of these models for real-time or edge deployment.

The results shown in Table 1 are averages for test files across seven SNR levels, spanning from -10 dB to 20 dB with an interval of 5 dB. Across six metrics, the proposed HF-DNN framework achieves a PESQ score of 2.53, surpassing MANNER, while remaining similar to the scores of the Reference DNN and MP-SENet. It can be seen that identical STOI values across HF-DNN and all competing methods suggest that all models effectively enhance intelligibility under noisy conditions. However, the proposed HF-DNN framework achieves the best SSNR of 3.45 dB, outperforming all other algorithms.

Table 1: Performance Comparison of Speech Enhancement Algorithms

Method	PESQ	STOI	SSNR	CSIG	CBAK	COVL	Para.	MAC/s
Noisy	1.96	0.71	-1.98	1.14	2.14	2.39	-	-
MP-SENet	2.85	0.87	2.46	3.39	2.61	2.99	2.05M	37.14G
SEMamba (Reference DNN)	2.88	0.87	2.48	3.40	2.63	3.01	2.2M	32.73G
MANNER	2.46	0.87	3.19	3.21	2.63	2.78	24.1M	7.853G
HF-DNN	2.53	0.87	3.45	3.25	2.62	2.77	71k	418M
Knowledge Distillation	2.52	0.87	2.86	3.10	2.58	2.77	21k	415M

The CSIG metric, which quantifies overall intelligibility and speech quality by evaluating distortions introduced by the enhancement process, shows that HF-DNN performs competitively. COVL assesses the subjective preference of listeners towards the enhanced speech, balancing between noise suppression and speech distortion. In CSIG and CBAK, the proposed framework provides almost similar performance to the Reference DNN and MP-SENet and performs better than MANNER.

CBAK measures the level of residual noise in the enhanced speech. Identical CBAK values across HF-DNN and all competing algorithms suggest that all models are equally effective in suppressing background noise.

With respect to Knowledge Distillation, the proposed framework outperforms Knowledge Distillation in SSNR and CSIG, while providing almost similar performance in other metrics.

4.6. Reduction in Computational Cost

The HF-DNN framework demonstrates a significant reduction in trainable parameters compared to MANNER, the Reference DNN and MP-SENet making it a much lighter model while maintaining competitive enhancement performance. When compared to MANNER, which consists of 24.1 million parameters, HF-DNN achieves an impressive 99.7% reduction in model size. This means that MANNER requires nearly 340 times more parameters than HF-DNN, showcasing the efficiency of the structured hierarchical framework of HF-DNN. The Reference DNN used in the experiments has a significantly larger model size, making HF-DNN a much more compact alternative with only 71K parameters, while still delivering similar speech enhancement performance with 96.8% reduction in trainable parameters. Compared to MP-SENet, which has 2.05 million trainable parameters, HF-DNN reduces the model complexity by 96.5%.

In terms of MAC operations, HF-DNN reduces computations by 98.9% compared to MP-SENet. This means that MP-SENet requires nearly 90 times more computational power, making HF-DNN significantly more efficient for deployment on edge devices and low-power environments. In comparison to SEMamba, HF-DNN reduces the computational cost by 98.7%, ensuring a much lower inference time while still delivering similar speech enhancement performance. Against MANNER, the HF-DNN framework achieves a 94.7% reduction, further emphasizing its computational efficiency.

The HF-DNN framework has slightly higher trainable parameters than Knowledge Distillation but almost similar MAC operations. One advantage of the HF-DNN framework is the flexibility in choosing DNN architectures for both DNN classifiers and speech enhancement DNNs, unlike Knowledge Distillation, which relies on a single DNN model for both teacher and student models. This may facilitate applicability of proposed model to diverse applications. It is also possible that HF-DNN

may also have an edge over single large DNNs in applications requiring DNN adaptation.

5. Conclusions

In this work, we introduce a novel Hierarchical Framework for DNN (HF-DNN) architectures for employing multiple simpler DNNs to overcome the high computational cost of a single large DNN model for speech enhancement. The proposed HF-DNN is based on the hierarchical clustering of the structured codebook of log power spectrum (LPS) vectors of clean speech data to categorize a set of input noisy LPS frames or segments of multiple frames into multi-layered classes. By identifying the most probable class to which an input belongs, a simpler speech enhancement DNN model can be utilized, thereby avoiding the need for a single, complex DNN for speech enhancement.

The reduction in complexity is more than 90% compared to the state-of-the-art reference DNNs. Moreover, this computational reduction is complemented by an improvement in speech enhancement performance in SSNR and comparable results in the other evaluation metrics. The HF-DNN can be utilized with any DNN model, and it can also be applied to speech processing tasks other than speech enhancement.

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